No-code software development · scoping deliverable

Application Assistant

A Discord bot that helps you build your CV & Motivation Letter based on the job description

A Discord bot that ingests your base CV once, reads any job link you send it, asks you targeted questions to fill your skill gaps, and generates a perfectly tailored application.

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Tool n8n **Channel** Discord **Database** n8n Data Tables **AI** GEMINI (openrouter) **Target** Level 3

In one sentence. Students who apply to many internships rewrite their CV and cover letter by hand for every posting, which is exhausting, error-prone, and takes hours.

What they do today. They duplicate a base CV and manually reword it. Generic AI tools exist, but they are tedious because they require the student to manually copy-paste their background, the job description, and the prompt back and forth for every single application.

What we bring ?. you interact exclusively with a private Discord bot:

1. **Setup:** you upload your CV (PDF) to the bot.
2. **Analyze:** Whenever I find a job I want, I just paste the URL into Discord. The bot scrapes the job, compares it to my saved CV, and asks me *specific questions* in the chat.
3. **Generate:** you reply, and the bot outputs a targeted CV, a matching Motivation Letter, and an Interview Cheat Sheet. (keep in mind your answers)

The Three Checks Validated:

- **Trigger I control:** The flow starts on demand via a Discord message in my private server.
- **Data I can reach:** Discord API, open web job postings (via HTTP fetch), and Groq for AI.
- **Boring failure:** If the bot fails or hallucinates, the worst outcome is a bad draft CV that I simply don't send. Zero risk of sending a bad application to a company.

1 The trigger and testing plan

Main trigger: Discord Trigger. It fires on every message sent to the private channel.

Testing on demand: send the bot a message, anytime. I keep a fixed set of test inputs to resend after every change to ensure stability:

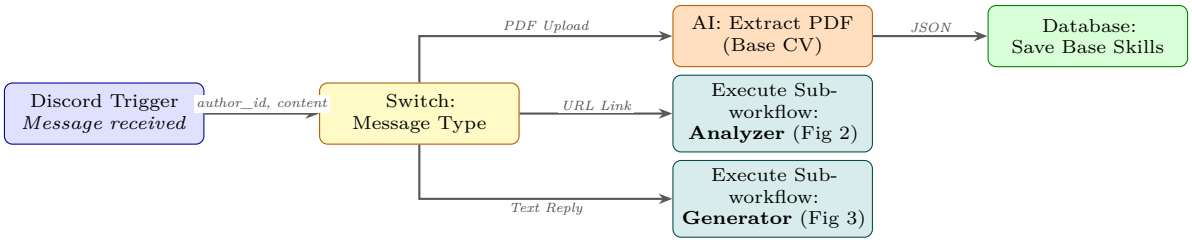
Test input	What it checks
PDF File Upload	Base CV extraction, AI parsing, and insertion into the database.
Link to an internship posting	URL fetching, AI gap analysis, state updating, and Discord question reply.
Text reply: "I used Agile during my grad project"	Skill extraction, database update, and final document generation.
Link to a blocked or JavaScript-only page	Fallback path that asks the user to paste the text instead of failing.
`/help`	A command gets the usage message instead of silence.

2 Architecture diagrams

Four workflows. Labels on the arrows show what travels between blocks.

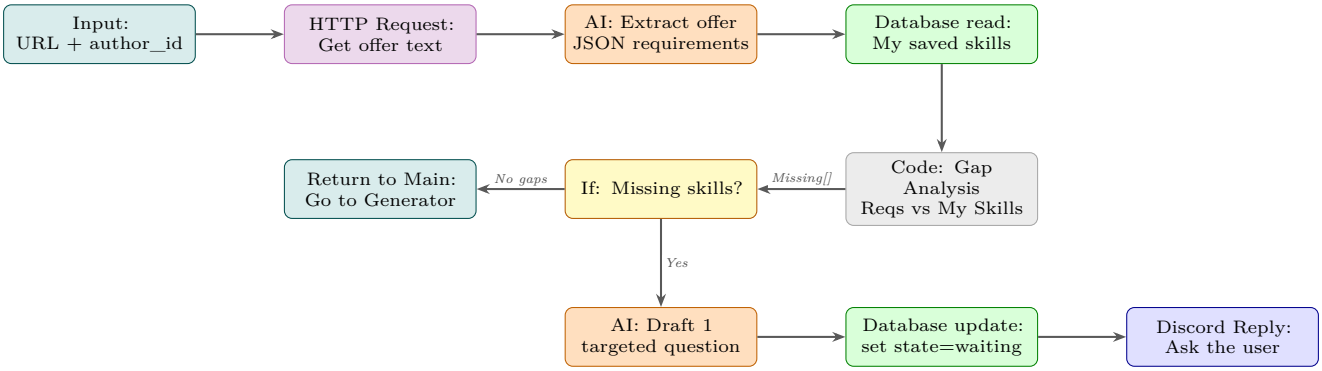
2.1 1. Main Workflow: Receiving a message

One Switch sorts every message based on the input type. It acts as a traffic controller, delegating complex tasks to specialized sub-workflows.



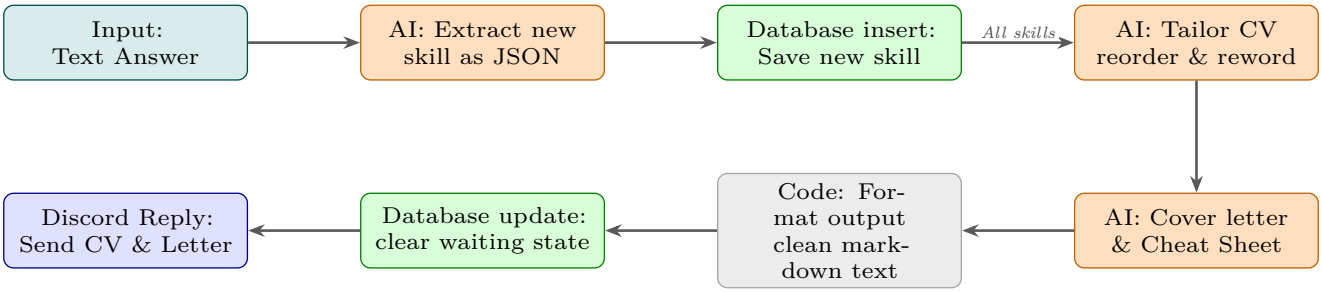
2.2 2. Sub-workflow: Analyzer (Finding the skill gaps)

Triggered by a URL. The bot fetches the job, reads your current skills, finds what you are missing, and asks you a question.



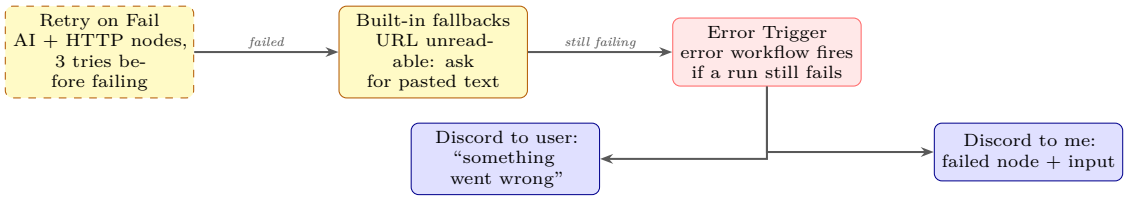
2.3 3. Sub-workflow: Generator (Creating the documents)

Triggered by your text reply. The bot saves your new skill permanently, writes the CV, Letter, and Cheat Sheet, and sends them.



2.4 4. Error Handling

AI and HTTP nodes retry 3 times. If a run still fails, I get the details and the user gets a clear message.



3 Level and cut line

Target: Level 3. The project branches (Switch), uses conversational state memory (pausing to wait for an answer), chains services, and heavily utilizes sub-workflows and robust retries.

What I cut first if I run out of time, in strict order:

1. **The Interview Cheat Sheet:** Stick to just the CV and Motivation letter.
2. **The Motivation Letter:** Just generate the CV.

3. **The Refining Question (Analyzer):** Bypass the gap analysis. The bot just takes the job link and generates the CV based purely on whatever is currently inside the Base CV database without asking clarifying questions.

4 Minimum technical requirements

Requirement	Where it is used
If or Set node	If: “Are there missing skills?”. Set: formatting the final Discord reply text.
Code node	Clean the fetched web page into text, string-match job requirements against database skills to find gaps.
Router or If/Switch	Switch on Discord message type (link, text answer, PDF upload).
API or MCP connection	HTTP Request to fetch the job posting web page. Discord API.
Message trigger	Discord Trigger (acting on new channel messages).
Database	n8n Data Tables: `facts` tab (skills, experiences) and `sessions` tab (tracking conversation state).
AI node	LLM Chains with structured JSON output for job extraction, plus LLM nodes for the CV, Letter, and Questions.
Sub-workflow	“Analyzer” and “Generator”, called cleanly from the main workflow.
Error handling and retries	Retry on Fail (3 tries) on AI and HTTP nodes, fallback branches, and an Error Workflow that notifies me.

5 Future possibilities of improvement

Once the core pipeline is stable, we can easily implement other functionalities such as:

- **Kanban Board Tracker:** Connect the database to a Notion or Google Sheets Kanban board that automatically moves cards from "To Apply" to "Applied" when the CV is generated.
- **Inbox Watcher:** Add an IMAP/Gmail trigger to scan my email for recruiter replies. If a recruiter asks for an interview, the bot reads the email, fetches the original job description from the database, and automatically sends me prep materials.
- **Automated Follow-ups:** A daily Schedule Trigger that checks the database for jobs applied to > 7 days ago and drafts a follow-up email.